

Supply Chain Performance and Delivery Risk Analysis Using Microsoft Excel and Power BI

1. Project Overview and Objective

This project focuses on analysing supply chain operations using Microsoft Excel and Power BI. The dataset contains sales transactions, customer information, product details, shipping records, and delivery performance metrics.

The primary objective is to transform raw supply chain data into meaningful business insights through data cleaning, exploratory data analysis (EDA), and interactive dashboard development.

- Analyse regional sales performance.
- Identify highly profitable product categories.
- Evaluate customer segment contribution to revenue.
- Assess shipping performance and delivery risks.
- Develop an interactive Power BI dashboard for business decision-making.

2. Data Sources

Source Description: Supply Chain Retail Dataset

Domain: Supply Chain Analytics / Retail Sales Analytics

Dataset Size: 19,999 Records and 37 Columns

Data Period: Historical sales and shipping transaction records. The dataset contains customer, product, sales, shipping, and delivery information that enables comprehensive analysis of business operations and logistics performance.

3. Problem Statement

- Identify high-performing regions.
- Identify profitable product categories.
- Analyse customer purchasing behaviour.
- Evaluate delivery delays and shipping risks.
- Improve operational efficiency through data-driven insights.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Order Id	Integer	Unique Order Identifier
Order Country	Text	Country of Order
Order Region	Text	Sales Region
Customer Segment	Text	Customer Category
Category Name	Text	Product Category
Sales	Decimal	Sales Revenue
Order Profit Per Order	Decimal	Profit Generated
Profit Margin %	Percentage	Profitability Percentage
Shipping Mode	Text	Delivery Method
Delivery Status	Text	Shipping Status
Late delivery risk	Integer	Late Delivery Indicator
Order Month	Text	Month of Order
Order Quarter	Text	Quarter of Order
Order Year	Integer	Year of Order

5. Tools & Technologies

Microsoft Excel

Data Cleaning, Duplicate Check, Missing Value Handling, Calculated Columns, Pivot Tables.

Microsoft Power BI

Data Modelling, DAX Measures, Dashboard Development, Interactive Reporting.

6. Data Pre-Processing

Tasks Performed:

Data Cleaning & Transformation: Removed duplicates, handled missing values, standardized formats, and created calculated fields.

- Delivery Delay
- Profit Margin %
- Order Quarter
- Order Month
- Order Year

Delivery delay ▼	Profit Margin % ▼	Order Quarter ▼	Order Month ▼	Order Year ▼
0	24.70%	Q3	September	2015
0	33.46%	Q3	September	2017
1	24.75%	Q2	May	2015
1	35.20%	Q3	July	2016
1	48.51%	Q4	November	2015

Filtering & Sorting: Organized data to focus on relevant records.

Pivot Tables: Generated Pivot Tables for data summarisation and initial insights.

- Profit by Category
- Sales by Region
- Sales by Customer Segment
- Shipping Mode Analysis
- Late Delivery Risk Analysis
- Delivery Status Analysis
- Monthly Sales Trend
- Top 10 Countries by Sales
- Profit Margin by Category

Sales by Region		Sales by Customer Segment		Profit by Category		Shipping Mode Analysis		
Order Region	Sum of Sales	Customer Segment	Sum of Sales	Category Name	Sum of Order Profit Per Order	Shipping Mode	Average of Days for shipping (real)	Sum of Sales
Western Europe	651590.0915	Consumer	2111247.17	Fishing	75983.12073	Standard Class	3.980564631	2435063.057
Central America	616374.9622	Corporate	1231596.324	Cleats	61350.28994	Second Class	3.966734903	794447.3749
South America	326553.8362	Home Office	714315.1038	Camping & Hiking	48296.37953	First Class	2	622576.6219
Northern Europe	241380.5647	Grand Total	4057158.598	Cardio Equipment	48204.90976	Same Day	0.490548204	205071.5439
Southeast Asia	228325.1743			Water Sports	40515.15991	Grand Total	3.486624331	4057158.598

Late Delivery Risk Analysis		Delivery Status Analysis		Monthly Sales Trend		Top 10 Countries by Sales		Profit Margin by Category	
Late Delivery Risk	Count of Order Id	Late Delivery Risk	Count of Order Id	Order month	Sum of Sales	Order month	Sum of Sales	Order month	Average of Profit Margin %
0	9071	Advance shipping	4692	January	382854.4172	Estados Unidos	533156.5708	Accessories	0.140179681
1	10928	Late delivery	10928	August	357104.537	Francia	324780.4957	As Seen on TV!	0.207304348
Grand Total	19999	Shipping canceled	874	May	356074.477	México	275024.4954	Baby	0.13155564
		Shipping on time	3505	March	351318.6967	Alemania	228555.2341	Baseball & Softball	0.094583497
		Grand Total	19999	September	347750.6461	Australia	183283.1234	Basketball	0.128553636

7. Data Modelling and DAX (Power BI)

Data Model:

Since the dataset was imported as a single fact table containing all required business attributes, no additional dimension tables or relationships were required.

Calculated Columns & DAX Measures:

Calculated Columns (Excel)

Calculated Column	Purpose	Formula Used
Profit Margin %	Measures profitability of each order	=[@ [Order Profit Per Order]]/[@Sales]
Order Month	Used for Monthly Sales Trend analysis	=TEXT ([@ [order date (Date Orders)]], "MMMM")
Order Quarter	Used for Quarterly filtering and analysis	="Q"&ROUNDUP(MONTH(AN2)/3,0)
Order Year	Used for year-wise analysis	=YEAR ([@ [order date (Date Orders)])]
Delivery Delay Days	Measures shipping delay duration	=[@ [Days for shipping (real)]]-[@ [Days for shipment (scheduled)]]

DAX Measures (Power BI)

Measure	Formula Purpose	Formula Used
Total Sales	Sum of Sales	Total Sales = SUM ('Supply Chain'[Sales])
Total Profit	Sum of Order Profit Per Order	Total Profit = SUM ('Supply Chain'[Order Profit Per Order])
Total Orders	Count of Order Id	Total Orders = COUNT ('Supply Chain'[Order Id])
Profit Margin %	Profit ÷ Sales	Profit Margin % = DIVIDE ([Total Profit], [Total Sales],0)
Late Orders	To find Late Delivery %	Late Orders = CALCULATE (COUNT ('Supply Chain' [Order Id]), 'Supply Chain'[Late delivery risk] = 1)
Late Delivery %	Late Orders ÷ Total Orders	Late Delivery % = DIVIDE ([Late Orders], [Total Orders],0)

8. Analysis and Visualizations (Power BI)

Dashboard Features:

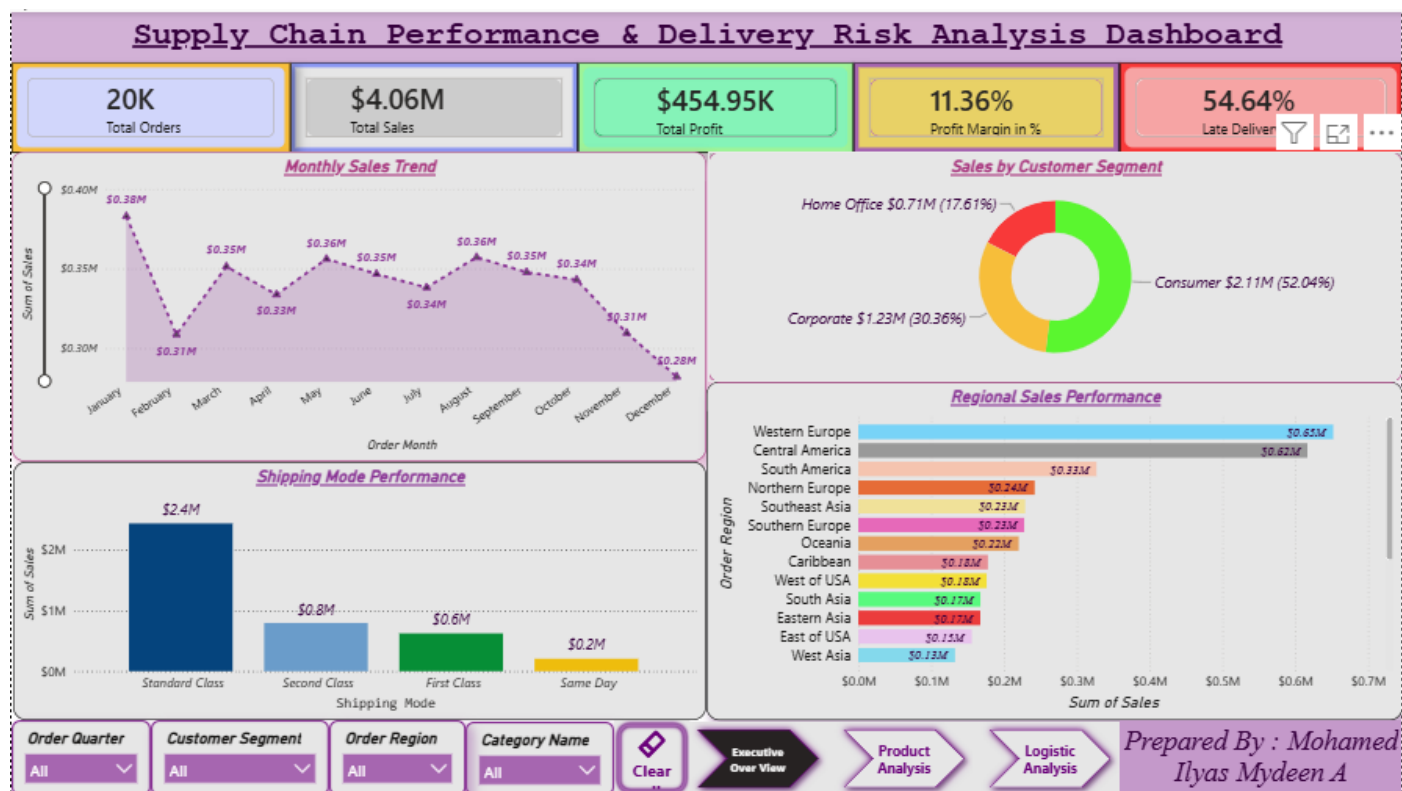
1. KPI Cards
 - Total Sales
 - Total Profit
 - Total Orders
 - Profit Margin %
 - Late Delivery %
2. Monthly Sales Trend (Line Chart)
3. Regional Sales Performance (Bar Chart)
4. Customer Segment Analysis (Donut Chart)
5. Top 10 Profitable Categories (Bar Chart)
6. Top Countries by Sales (Bar Chart)
7. Shipping Mode Analysis (Column Chart)
8. Delivery Status Analysis (Donut Chart)
9. Regional Performance Summary (Table)
10. Geographical Sales Analysis (Drill Down)

Features:

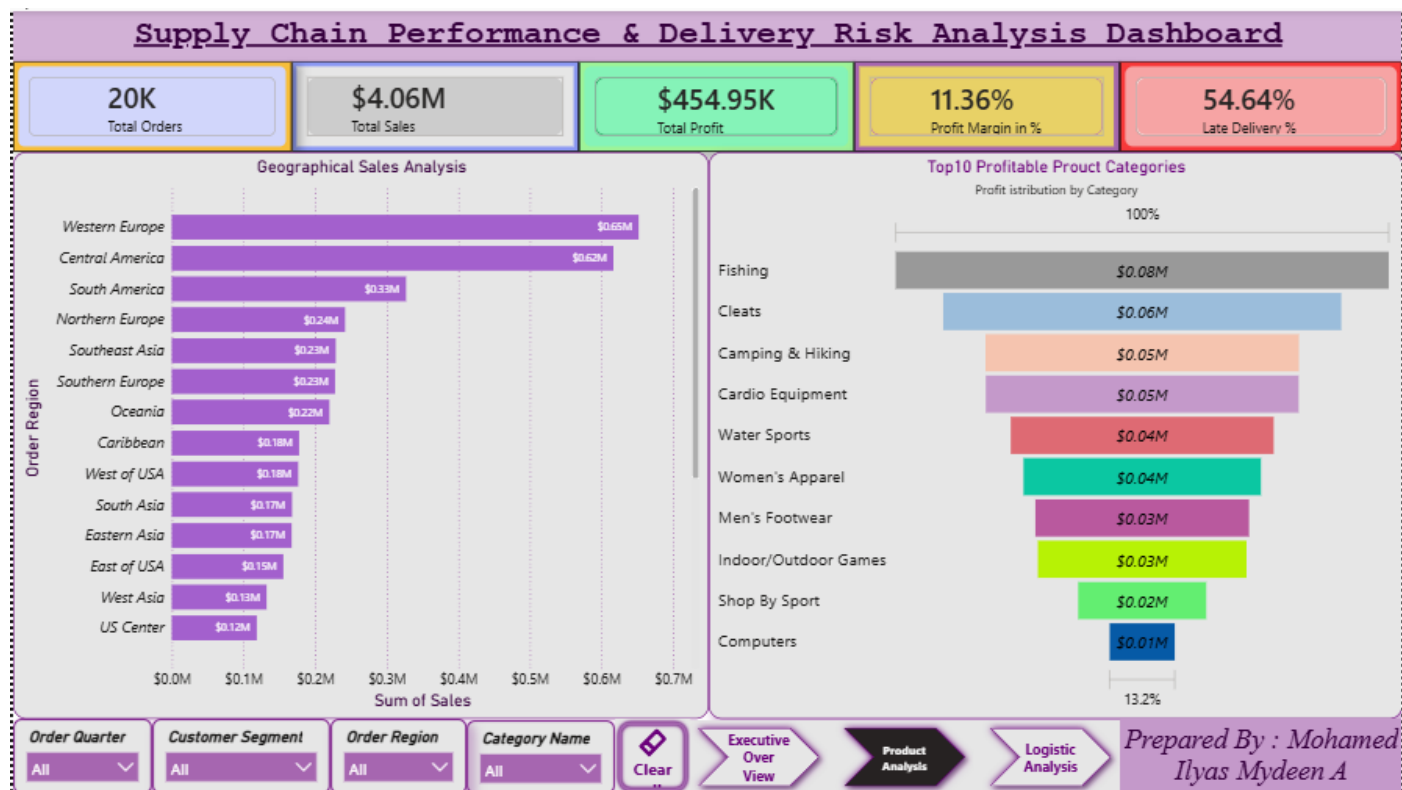
- Slicers
- Filters
- Drill-down
- Bookmarks
- Conditional Formatting

Report /Dashboard

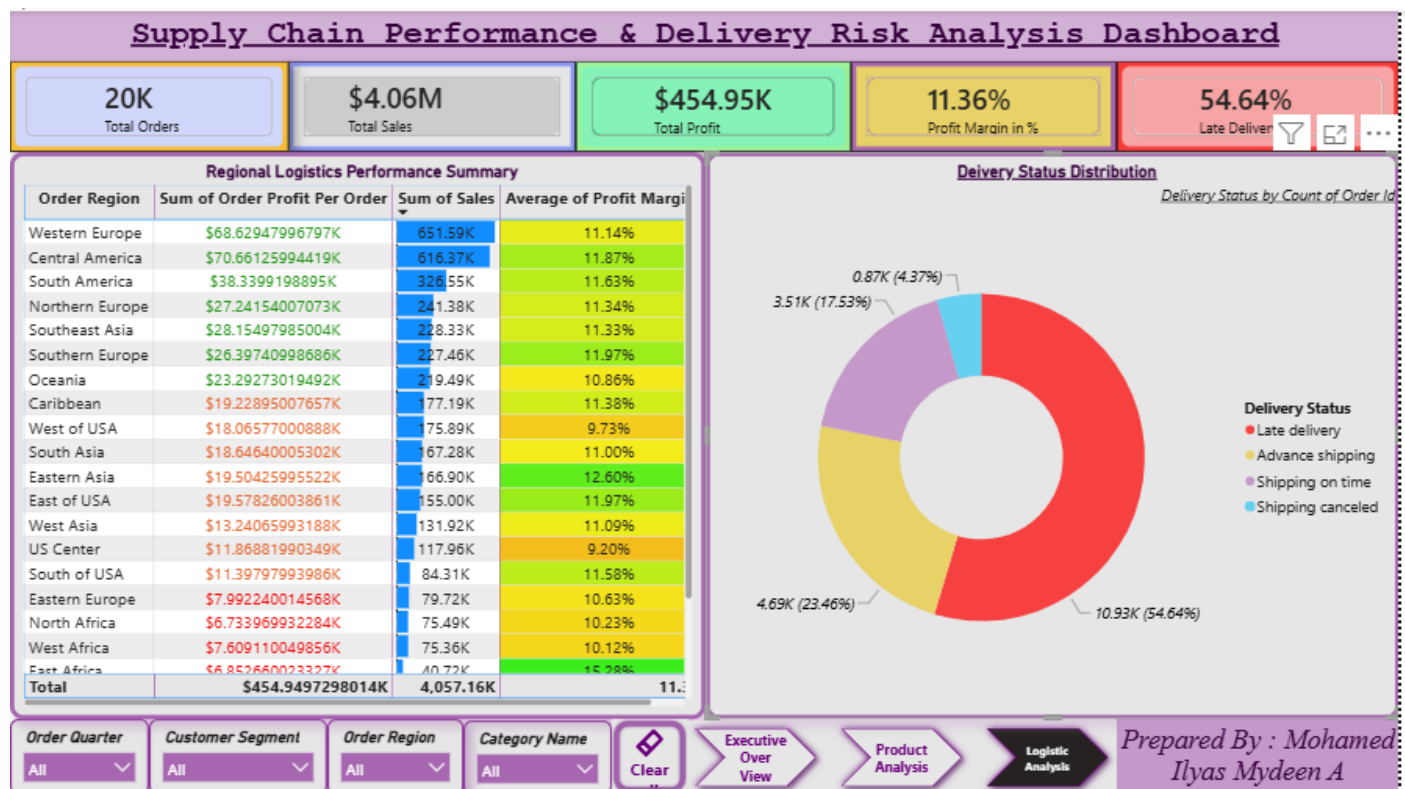
Executive Overview



Product Analysis



Logistic Analysis



9. Insights & Conclusions

Key Insights for Report

Executive Overview

- Total Sales reached **4.06 million**.
- Total Profit reached **454.95 thousand**.
- Consumer segment contributed over **52% of sales**.
- Western Europe generated the highest revenue.
- Standard Class generated the highest sales volume.

Product Analysis

- Fishing category generated the highest profit.
- Cleats and Camping & Hiking were also major contributors.
- Product profitability is highly concentrated in the top-performing categories, with Fishing generating the highest profit contribution.

Logistics Analysis

- Late deliveries account for **54.64%** of total orders.
- Standard Class accounted for approximately **60%** of total sales, indicating customer preference for cost-effective shipping methods.
- Significant opportunity exists to improve logistics efficiency.

Descriptive Insights

- Western Europe generated the highest sales.
- Consumer segment contributed the largest revenue share.
- Standard Class accounted for the highest sales volume.

Diagnostic Insights

- More than 50% of orders experienced delivery delays.
- Sales and profits are concentrated in a limited number of product categories, with **Fishing, Cleats, and Camping & Hiking** contributing significantly to profitability.

Predictive Insights

- Improving logistics performance can reduce late delivery risk.
- High-performing regions can drive future revenue growth.

Prescriptive Insights

- Improve logistics efficiency to reduce delivery delays.
- Prioritize inventory for high-profit categories.
- Focus marketing efforts on high-performing regions.
- Optimize shipping methods to improve customer satisfaction.

10. Conclusions

The project successfully transformed raw supply chain data into actionable business insights using Excel and Power BI. Through data cleaning, exploratory analysis, and dashboard development, key trends related to sales performance, profitability, customer behaviour, and logistics operations were identified. The interactive dashboard enables decision-makers to monitor business performance and make data-driven decisions for improving operational efficiency and profitability.

Submitted By:

Mohamed Ilyas Mydeen A